

Mathematical Foundations of the BRICS Geopolitics Simulator

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1 Introduction

Mathematical modeling of geopolitics seeks to formalize strategic interaction, feedback, and uncertainty within international systems. States are treated as strategic agents embedded in coupled dynamic systems, where decisions influence both contemporaneous payoffs and future system states¹. This abstraction enables reproducible scenario analysis and stress testing under controlled perturbations.

2 Data & Signals

Let $x_{i,t}$ denote a macroeconomic indicator for state i at time t , such as GDP growth or trade balance. News-derived signals are extracted from text corpora using term-frequency or sentiment-weighted counts². A generic signal vector is

$$\mathbf{s}_{i,t} = \left(x_{i,t}^{(1)}, \dots, x_{i,t}^{(k)}, \nu_{i,t} \right), \quad (1)$$

where $\nu_{i,t}$ aggregates normalized news intensity.

3 Game-Theoretic Models

3.1 Adaptive Tariff Game

Each state chooses a tariff level $a_i \in [0, 1]$. Payoffs are given by

$$\pi_i(a_i, a_{-i}) = \alpha_i a_i - \beta_i a_i a_{-i} - \gamma_i a_i^2. \quad (2)$$

Strategies update via a logistic response:

$$a_i^{(t+1)} = \frac{1}{1 + \exp\left(-\lambda \pi_i^{(t)}\right)}. \quad (3)$$

Logistic adjustment ensures bounded rationality and smooth adaptation³.

¹Formal strategic interaction is standard in game theory; see [6].

²Domain-specific lexicons for political and financial text are widely used; see [3, 4].

³Logistic response functions originate in Verhulst-type growth and learning models; see [9].

3.2 Energy & Currency Coordination Game

Let $C \subseteq N$ be a coalition. Coalition payoff is

$$U(C) = \sum_{i \in C} u_i - \delta |C|^2, \quad (4)$$

where $\delta > 0$ captures coordination costs. Marginal benefits are nonlinear due to scale effects, consistent with coalition formation theory⁴.

3.3 Richardson-style Tariff Arms Race

Tariff escalation is modeled in continuous time:

$$\frac{dx}{dt} = ay - bx + r, \quad (5)$$

$$\frac{dy}{dt} = cx - dy + s. \quad (6)$$

Stability depends on the eigenvalues of the coefficient matrix. If $ac < bd$, the system converges to equilibrium; otherwise divergence occurs⁵.

4 Geopolitical Knowledge Graph

States are nodes; weighted edges w_{ij} encode alignment or pressure. The adjacency matrix W_t evolves as

$$W_{t+1} = W_t + \eta f(\mathbf{s}_t), \quad (7)$$

where $f(\cdot)$ maps news signals to edge updates⁶.

5 Scenario Risk Model

Define chokepoint exposure $\kappa_i \in [0, 1]$. Shock propagation obeys

$$\mathbf{r}_{t+1} = A\mathbf{r}_t + \boldsymbol{\epsilon}_t, \quad (8)$$

with systemic risk

$$R = \sum_i \omega_i r_i. \quad (9)$$

⁴Coalitional stability and externalities are treated in [1, 7].

⁵This structure follows Richardson's arms race model; see [8].

⁶Dynamic network modeling is standard in network science; see [5].

6 Model Limitations & Failure Modes

Limitations include non-stationary behavior, simplified payoff structures, biased data, parameter sensitivity, and the tension between deterministic and stochastic representations⁷.

7 Conclusion

The simulator integrates game theory, dynamical systems, and network models into a coherent mathematical architecture while remaining subject to interpretive constraints.

⁷See discussions in political science modeling [2].

References

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